# Chapter 14 exercises • Protection

## OS14-E01 • Domains and least privilege

Phase A needs only read(S); B only write(T); C parses untrusted input.

1. What excess does a static union domain expose? Define dynamic restrictions.
2. Does user mode alone isolate these phases?

## OS14-E02 • Exact matrix updates

Initially A[D1, F]={r*, owner}, A[D2, F]={w}, A[D3, F]=∅; D1 also holds control(D2). Here r* includes read and its delegation right. Limited copy gives r in the same object column while retaining the source; transfer removes both r and delegation authority from the source and gives them to the recipient. owner(F) may add or remove rights in F's column; control(D2) may only remove rights in D2's row.

1. Limited-copy r to D3, then transfer r* to D2. Show F's column after each.
2. Which two rights can remove D2's w? Can control add x to D2?

## OS14-E03 • Reachable domains versus simultaneous rights

D1→D2→D3; only D3 can print P. Current domain is D1.

1. Can it print immediately? Can it print later legally?
2. Delete D2→D3 with no other path. Prove printing impossible for every subsequent switch-only trace.

## OS14-E04 • ACLs, capabilities, and existing opens

F ACL=[U1:r, U2:rw], G ACL=[U2:r, U3:w], allow-only/default-deny. Assume a kernel can issue and protect unforgeable capabilities encoding this relation, and holders cannot enlarge rights; transposing the table alone provides no such protection.

1. Convert to user capability rows.
2. U1 already has one read-only open handle. Delete U1's ACL entry and consider only this handle. Compare write and read under (a) ACL checking only on open and a still-live handle and (b) per-use generation validation with this generation revoked.

## OS14-E05 • Generation numbers and ABA

Cap=(5, g3, r), table[5]=(F, g3, live). Revoke F and reuse slot 5 for G. On each reuse increment generation: unbounded in part 1, modulo 4 in part 2.

1. Explain slot-only failure and rejection with generations.
2. With a two-bit counter and an everlasting old token, after how many reuse steps does g3 recur? Give a remedy.

## OS14-E06 • Active RBAC roles

U is assigned Backup and Grade. They cannot be active in the same session. Backup grants readSubmissions; Grade grants writeGrades.

1. Evaluate writeGrades under active={Backup}. What must switching check?
2. Does another session with Grade violate the stated rule?

## OS14-E07 • Historical Java inspection

SE 8, no explicit context or limited permissions: stack N→L→A has p at N/L but not A; inherited context holds p.

1. Evaluate ordinary inspection and L doPrivileged.
2. If L lacks p, does its privileged boundary help? Separately, suppose N, L, A have p but inherited context lacks p. Decide both with and without L's privileged boundary.
3. What fails when L accepts arbitrary addr and connects in a privileged block?

## OS14-E08 • Delegation and information flow

D1 holds F:r and G:w, but neither F:r* nor owner(F). Only an F:r* holder can delegate F rights; no other grant or switch is available.

1. Can it grant D2 read authority? Can it copy F's data into G?
2. Why does attenuation induction not directly apply to a system with owner rights?

## References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz et al., Operating System Concepts, 9th ed., Ch. 14. Oracle: Java SE 8 AccessController. Oracle: Solaris 10 RBAC.

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https://github.com/xinhi6716/computer-science-self-study